From The Frontiers of Cognitive Psychology

Recent cognitive research at MIT has produced, among several other gizmos, the Galvactivator—a glove-like wearable device that senses the wearer's skin conductivity, and maps these onto a LED display. When there is an arousal of any kind in the person's body, the Galvactivator display glows brightly. The Galvactivator has many potentially useful purposes, ranging from self-feedback for stress management, to facilitating conversation between two people, to new ways of visualising excitement levels in performance situations. Visit ***http://affect.media.mit. edu/*** for more details.

king around. But there was a very advanced chess-playing program that actually took the rook, and opened up the wall of pawns!

One of the most obvious difficulties in building smart machines is that computers lack common sense. We acquire common sense through years of learning, and it is not an easy task to transfer this to our machines.

But, what about learning? Can a machine not learn, and become intelligent? Yes and no. We learn mostly by analogy; we tell a child that ten times two is twenty—"just add a zero to the two"; and that ten times three is thirty—"just add a zero", again. The child will probably figure that ten times four is forty, by adding a zero—without being told this, but machines don't seem to be able to learn this way.

Machines can learn, and that's where ANNs come in. ANNs learn by interacting with a teacher, and they are built by training them in the following manner: When a question is put to the ANN, and it gives the wrong answer, its input is changed to reflect the deviation of its answer from the correct one, and so on, until it outputs the correct answer. For example, an ANN can be trained to find whether a sentence is grammatically correct, or not. But it also turns out that intelligent behaviour is better exhibited by computers by training them according to rule-based systems.

Both the ANN approach and the regular or symbolic approach, go into the field called AI. Again, AI means many different things: a science, a philosophy, a range of technologies, a range of beliefs, a field of engineering—and more.

The AI philosophy

As a philosophy, AI is the idea that a machine can do whatever any brain can do. But this is one version of it, called strong AI. Weak AI, as the name suggests, says that a machine can do quite a lot of what a brain can do. Operational AI says that ...well, there are lots of flavours of AI: to each his own belief. Minsky said that brains were machines made of meat; this statement almost defines Strong AI. The Strong AI viewpoint is AI at its most extreme: it says that we are machines, and predicts that everything you see in any sci-fi movie that involves robots, and more, is possible. Minsky's dream involves one of the biggest AI controversies, and in all of science—as well as in popular-science culture: whether we can ever pass on our intelligence to our pet robots.

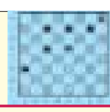
You can try and build machines that just behave in an intelligent manner, or you can try and find ways to build one that actually is intelligent, and understands what it is doing. To a Strong AI practitioner, these two are essentially the same thing—if a machine behaved intelligently, there was no way you could say it did not understand what it was doing. And to adherents of the viewpoint that human intelligence can never, even in theory, be enacted on a machine, it was very important indeed whether a machine actually understood what it was doing. We could say that most people working on AI

Robot Soccer

RoboCup (***www.robocup.org***) is the Robot Soccer Competition—a soccer competition for, well, robots. There are several leagues—the humanoid league, that is, actual moving human-like robots; the simulation league—a championship which looks like soccer computer games; the Sony Legged Robot League,i.e., teams of Sony Aibo robots; and others.

B-Soccer is a process that its creators say 'should' automatically produce soccer playing robots. It uses a simulation of soccer playing robot teams. The robots use ANNs in conjunction with a genetic algorithm—this is known as neuro-evolution—to learn how to play, and there is no control mechanism *per se*. The creators of B-Soccer are waiting to see if this kind of training will produce a winning team, or even a properly working team. Log on to ***http://www.robocup.org*** for further details.

What happened when




1943 McCulloch and Pitts publish seminal paper on neural networks

1950 The Turing Test is created, which challenges a machine to pretend to be human

1955 The term 'Artificial intelligence' is coined at the Dartmouth Conference

1959 Arthur Samuel's Chequers program wins against the best human players, including Robert Nealy

1962 The first industrial robots are created

1964 Bobrow's 'Student' program that can solve algebra problems developed

1965 Herbert Simon says, "by 1985, machines will be capable of doing any work a man can do"

1965 The DENDRAL project, that helps determine molecular structure, begins

1966 'Eliza', a program that mimics a psychiatrist, written

1967 The 'Perceptrons' paper ends funding for neural nets

1970 'Shrldu' program developed; the PROLOG language created

1972 Stanford's robotics funding cancelled by DARPA

1973 AI funding in the UK ends after the Lighthill report